The syntax is `\u{ORDINAL}` where `ORDINAL` is the hexadecimal ordinal value of the character, which means the character 合 is codepoint *U+5408*, which is the 21512th character in the code page. More details available here: https://www.unicode.org/cgi-bin/GetUnihanData.pl?codepoint=5408.

There are also two `mbstring` extension functions that are useful for moving between Unicode codepoint values and the associated characters. `mb_ord()` will give the decimal ordinal value of a character, and `mb_chr()` accepts an integer decimal codepoint value and will return the character:

```
var_dump(mb_ord("\u{5408}"));
var_dump(mb_chr(21512));
```

```
Output:
int(21512)
string(3) "合"
```

There are many other `mbstring` extension functions for handling multi-byte strings (of a variety of encodings, not just UTF-8); this chapter will make extensive use of them.

Transformation Formats

The original specification for Unicode also encompassed the mechanism for transferring data between systems, in that all 65,536 codepoints in the original design were to be represented by two bytes (16-bit). While Unicode was being worked on by Apple and Xerox, the ISO was also independently developing the parallel standard of ISO-10646 (known as the Universal Coded Character Set, or UCS), and the two groups agreed to synchronize their efforts to ensure the two code pages would be compatible. Both used a fixed two-byte (16-bit) code unit, which came to be known as UCS-2.

Both the Unicode Consortium (a non-profit organization incorporated in 1991 by the teams that had been working on the Unicode specification) and the ISO quickly realized that even 16-bit code units were insufficient to achieve the goal of providing for all the characters and symbols of human language. A four-byte system (UCS-4) was proposed in one of the drafts of ISO-10646; the Unicode consortium resisted this fixed-width system on the grounds that even on comparatively modern systems this would result in wasted storage, memory, and network bandwidth. The character encoding form that came to be known as UTF-16 was created as an alternative.